

Multi-person tracking for virtual reality surrounding awareness

Ayman Mukhaimar

Victoria University, Australia

Ang Yang

Victoria University, Australia

Yuan Miao*

Victoria University, Australia

Jun Zhao

Victoria University, Australia

Zora Vrcelj

Victoria University, Australia

Malindu Sandanayake

Victoria University, Australia

Bruce Gu

Victoria University, Australia

Melissa Chan

Victoria University, Australia

ABSTRACT

Virtual reality devices are designed to cover our vision so we are unaware of our surroundings and completely emerged into a different world. This limits VR user's ability to move freely in crowded areas, such as classrooms, which limits the user interactions in VR. We present a framework that enables VR users to see other people around them inside the VR environment with the help of an external 3D depth camera. The proposed framework enables multi-VR users to use one tracking camera and provides a cost-effective solution. The proposed framework can also help in performing collaborative activities.

Index Terms: Surrounding awareness in virtual reality—Virtual reality people tracking—Virtual reality skeleton tracking—Virtual reality real world interaction—Real world visualization in Virtual reality

1 INTRODUCTION

One of the promising technologies that will revolutionize the human-computer interface and user interaction with the digital world is virtual reality (VR). This next-generation immersive device will shape the future of education, entertainment, and how businesses operate. Recent VR devices are becoming stand-alone mobile platforms that require no external sensors, easy to set up, and are user-friendly. One of the drawbacks of using VR devices is that users lose sight of their surroundings as VR completely take over their vision. In addition, users get the feeling of being transported to a different world, which limits their ability to know what is happening around them in the real world. This could cause challenges in crowded scenarios such as classrooms, workshops, and gaming rooms or lead to VR breakdowns [2]. In such scenarios, VR users can easily collide with other people within their playing area/Guardian, which might lead to equipment damage or physical injuries.

To address this issue, we propose a framework where VR users will possess the ability to see other people around them in the virtual world. This is achieved using an external 3D tracking camera that can track all people in the room and send the tracking information to VR headsets. Tracking information is then used to show the VR user other people's locations as seen in Figure 1, or alert him if someone is in proximity. Different mechanisms can achieve this, i.e., a 2D map in the corner of the screen showing people's position or displaying an avatar at a location provided by the 3D camera (The avatar is controlled by the actual people's skeleton recorded by the 3D camera). The tracking camera not only can track people but also can track other moving objects surrounding the VR user, such as cats, dogs, etc. This way, VR users will not lose the sense of immersion and can realize and interact with real-world objects at the same time.

We also believe that our proposed framework can help in better engagement, interface, and learning content in collaborative activi-

*e-mail: Yuan.Miao@vu.edu.au

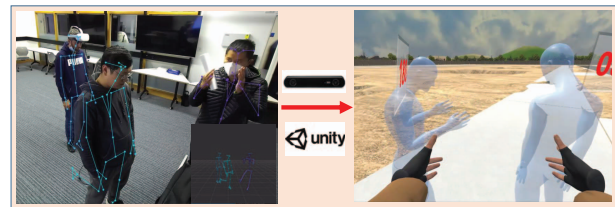


Figure 1: Surrounding awareness in virtual reality (VR), VR Users can see other people around in the actual world inside the VR scene.

ties. In scenarios where VR users are being observed and instructed by others [1,3], the ability to see instructors and their gestures inside the VR is more effective than only listening to voice instructions.

There have been some successful approaches for people's full-body tracking in VR [4–6]. Unlike those approaches, our framework is designed to enable multi-VR users to use one tracking camera, thus providing a cheaper solution than mounting tracking cameras over each VR device. In addition, our used 3D tracking camera is a stand-alone system that does not interfere with the VR device, can only send tracking information throughout the wireless network, and can be placed anywhere in the room as long as the VR user is within the tracking range of the 3D camera. Thus the 3D tracking camera can be turned on/off without influencing the performance of the VR device. This way, VR devices still keep their mobility and the advantage of being stand-alone platforms. Additionally, the tracking camera sends the skeleton information of everyone in the room, including the VR user's skeleton. Thus, we propose a matching algorithm that determines the user's skeleton and aligns the camera's coordinates with user coordinates in VR to correctly locate other people's positions in VR.

3D tracking cameras allow VR users to see other people with twofold benefits. Firstly, it helps in the spawn location of other VR users for multi-player scenarios, and secondly, the 3D tracking camera provides full-body tracking without requiring additional body sensors.

2 PROPOSED FRAMEWORK

Two versions of the VR program were designed: one for the server, which had the tracking camera, and one for the clients, which are the VR devices. We used the ZED 2 stereo camera¹ and its unity package for tracking. Once the server program starts, the program detects people with the help of the camera that would generate their skeletons. The program then generates avatars for every tracked person in the scene/game. The skeleton's information is then used to control the generated avatar's motion. The second version of the program, which is installed on the VR devices, receives the skeletons/avatars' information from the server and displays the avatar in the VR, as shown in Figure 2. The user can then see the avatars of the tracked people in the classroom, including his avatar. The avatars are displayed in the scene in positions determined by the

¹<https://www.stereolabs.com/zed-2/>

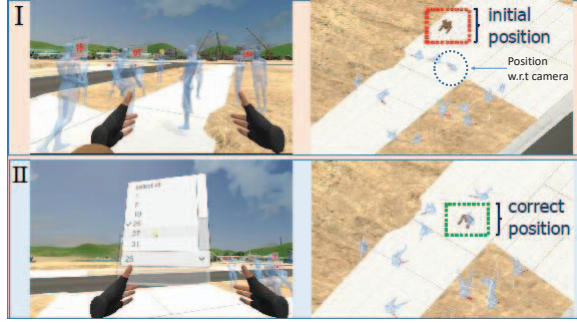


Figure 2: Manual alignment setup: (I) The VR user sees avatars of everyone being tracked by the camera including himself. The initial spawn position is shown in the red rectangle. (II) the VR user identifies which avatar is his. The user then selects that avatar ID and the user location is then changed to the avatar location as seen in the green rectangle.

server. Thus a calibration is needed to move the avatars from the camera/coordinates to the VR device coordinate. In order to do the calibration, we need to identify which avatar among the tracked avatars is the user avatar.

We propose two methods to identify the user avatar. The first one is manual selection. Manual avatar selection requires the user to select which avatar among the list of avatars is his. The user must do this only once at the beginning of the scene/game. After the avatar is selected, the two coordinates are calibrated. There are two drawbacks to this simple solution; the first one is that the camera is not able to accurately provide people's position, which could be due to non-ideal lighting conditions. Any slight error in people's position could increase over a long tracking period. Moreover, every time the user is occluded, the camera identifies the user as a new person who entered the scene, and loses track of the user.

Based on the drawbacks of the previous methods, we propose the second solution that auto-corrects the error in position every certain period of time using a matching algorithm that contiguously identifies the user based on his motion. The algorithm compares the user's actions with all avatar actions and matches the user with the avatar that has similar actions. Other features can also be used in the matching, such as the user's last known location, user height, etc. In order to match the head and hand positions, we first design a feature vector for every skeleton that is based on the distance between the three points (the head and the two hands) as seen in Figure 3 and the equation below:

$$Dn = [D1, D2, D3] \quad (1)$$

Where Dn is the feature vector for skeleton n that consists of the distance between the head, left hand, and right hand, as shown in Figure 3. We then find the feature vector H for the VR headset and the two hand controllers similarly. Based on these distances, we are able to find the label of the user skeleton as follows:

$$L_n = \min(|\hat{D}_n - \hat{H}|) \quad (2)$$

Where L_n is the label of the skeleton n that has the minimum difference between the normalized vectors D and H . The above simple classifier determines which skeleton belongs to the VR user instantly for every frame. This matching step is only required once the program starts or the user label changes (when the camera loses track of the user due to, i.e., occlusion). This way, the user label is always known, and we are able to update other avatars' locations with respect to the user location continuously. To validate the above classifier, we randomly gathered up to 20 skeletons from the recorded videos and compared those with the VR headset and controller

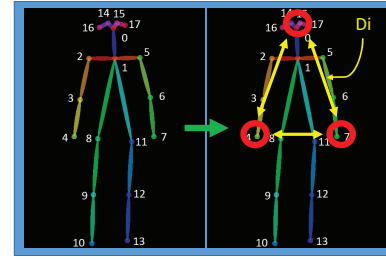


Figure 3: Keypoints of a tracked body model. D is the distance (D) in the 3D space between the head and the two hands.

positions. We compared over 2000 frames, and this simple classifier was able to achieve 90% accuracy over the 2000 cases. The classifier was enough to accurately identify the user among other people based on only hands and head postures. Future work will include more data to train a more robust classifier, i.e., multinomial logistic regression or a neural network, to recognize humans' behaviors and achieve better matching.

3 CASE STUDY

The above framework was put to the test in one of the university classes. We designed a VR scenario that showed some interactive content to the students as seen in Figure 1. The VR users were able to see other people in the classroom inside the VR environment. The classroom had good lighting conditions, the manual selection was used, and the result showed that everyone was tracked accurately.

4 CONCLUSION

We presented a virtual reality surrounding awareness framework that enables VR users to see other people around them inside the VR environment. The framework tracks people using an external 3D depth camera. The proposed framework was designed to enable multi-VR users to use one tracking camera to provide a cost-effective solution. Other moving objects, i.e., dogs, and cats, can be tracked similarly. The framework was successfully tested in a classroom, and VR users were able to see other people around them without losing the immersion feeling in VR.

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